

3. 真值表如下

A	B	C	F ₁	F ₂
0	0	0	0	0
0	0	1	0	0
0	1	0	0	0
0	1	1	0	1
1	0	0	0	0
1	0	1	1	1
1	1	0	1	1
1	1	1	1	1

当 $B=C=0$ 时 $F_1=0$, 其余 $F_1=A$

当 A, B, C 中有至少两个为 1 时 $F_2=1$, 其余 $F_2=0$

7. $Y_3 = X_3$

$$Y_2 = X_2 \oplus X_3$$

$$Y_1 = X_1 \oplus (MX_2 + \overline{M}Y_2)$$

$$Y_0 = X_0 \oplus (MX_1 + \overline{M}Y_1)$$

$M=1$ 时, $Y_3 = X_3$,

$$Y_2 = X_2 \oplus X_3$$

$$Y_1 = X_1 \oplus X_2$$

$$Y_0 = X_0 \oplus X_1, \text{ 显示循环码}$$

$M=0$ 时, $Y_3 = X_3$

$$Y_2 = X_2 \oplus X_3$$

$$Y_1 = X_1 \oplus X_2 \oplus X_3$$

$$Y_0 = X_0 \oplus X_1 \oplus X_2 \oplus X_3, \text{ 显示二进制码}$$

$\therefore M=1$, 从二进制码转换为循环码;

$M=0$, 从循环码转换为二进制码

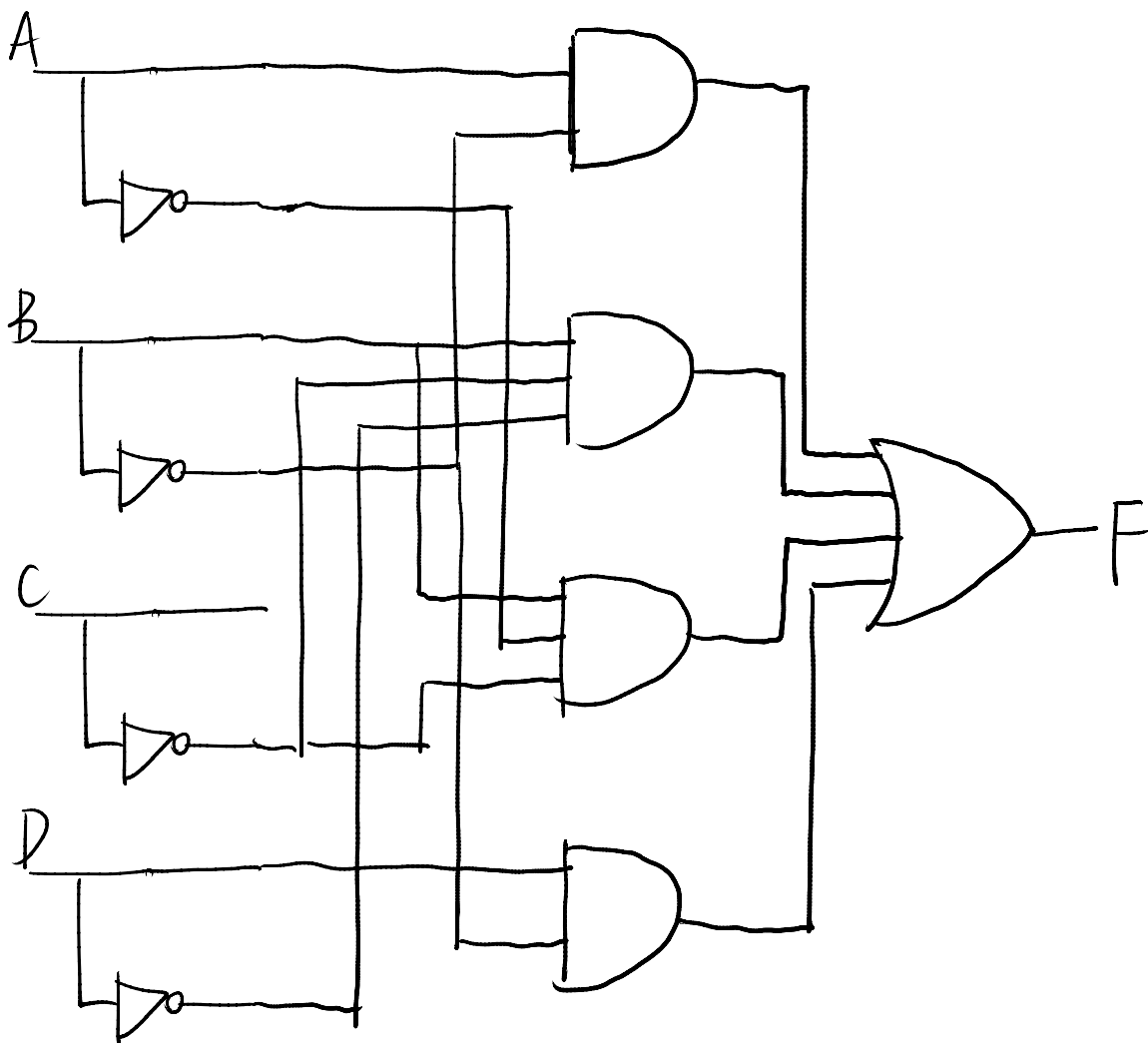
8.

A	B	C	D	F
0	0	0	0	0
0	0	0	1	1
0	0	1	0	0
0	0	1	1	1
0	1	0	0	1
0	1	0	1	1
0	1	1	0	0
0	1	1	1	0
1	0	0	0	1
1	0	0	1	1
1	0	1	0	1
1	0	1	1	1
1	1	0	0	1
1	1	0	1	0
1	1	1	0	0
1	1	1	1	0

画出卡诺图

CD \ AB	AB			
	00	01	11	10
00		1	1	1
01	1	1		1
11	1			1
10				1

$$\therefore F = \overline{B}D + A\overline{B} + B\overline{C}\overline{D} + \overline{A}B\overline{C}$$



15. 本题实际上是求二进制码和BCD码的转换
 设输入为 $Y_3 - Y_0$, 进位 C_0
 输出为 $S_3 - S_0$, 进位 C_1

Y_3	Y_2	Y_1	Y_0	C_0	S_3	S_2	S_1	S_0	C_1
0	0	0	0	0	0	0	0	0	0
		$\vdots$		$\vdots$			$\vdots$		$\vdots$
1	0	0	1	0	1	0	0	1	0
1	0	1	0	0	0	0	0	0	1
		$\vdots$		$\vdots$			$\vdots$		$\vdots$
1	1	1	1	0	0	1	0	1	1
0	0	0	0	1	0	1	1	0	1
		$\vdots$		$\vdots$			$\vdots$		$\vdots$

当 $C_0 = 0$. 输出 $S_3 S_2 S_1 S_0$
 C_1 (如省略)

$Y_3 Y_2$ $Y_1 Y_0$	00	01	11	10
00	0000	0100	0010 	1000
01	0001	0101	0011 	1001
11	0011	0111	0101 	0001
10	0010	0110	0100 	0000

当 $C_0 = 1$. 输出 $S_3 S_2 S_1 S_0$
 C_1 (如省略)

$Y_3 Y_2$ $Y_1 Y_0$	00	01	11	10
00	0110 	Φ	Φ	Φ
01	0111 	Φ	Φ	Φ
11	1001 	Φ	Φ	Φ
10	1000 	Φ	Φ	Φ

化简得 $S_3 = Y_1 C_0 + Y_3 \bar{Y}_2 \bar{Y}_1 \bar{C}_0 = Y_1 C_0 + Y_3 \bar{Y}_2 \bar{C}_0$

$$S_2 = \bar{Y}_1 C_0 + \bar{Y}_3 Y_2 \bar{C}_0 + Y_2 Y_1 \bar{C}_0 = \bar{Y}_1 C_0 + \bar{Y}_3 Y_2 + Y_2 Y_1$$

$$S_1 = \bar{Y}_1 C_0 + Y_3 Y_2 \bar{Y}_1 \bar{C}_0 + \bar{Y}_3 Y_1 \bar{C}_0 = \bar{Y}_1 C_0 + Y_3 Y_2 \bar{Y}_1 + \bar{Y}_3 Y_1$$

$$S_0 = Y_0 C_0 + Y_0 \bar{C}_0 = Y_0$$

$$C_1 = C_0 + Y_3 Y_2 \bar{C}_0 + Y_3 Y_1 \bar{C}_0 = C_0 + Y_3 Y_2 + Y_3 Y_1$$

18. 用 $I_1 I_2$ 表示供血者, $I_3 I_4$ 表示受血者

00、01、10、11 分别表示 A、B、AB、O 型血

F_1 、 F_2 表示绿灯 红灯

I_1	I_2	I_3	I_4	F_1	F_2
0	0	0	0	1	0
0	0	0	1	0	1
0	0	1	0	1	0
0	0	1	1	0	1
0	1	0	0	0	1
0	1	0	1	1	0
0	1	1	0	1	0
0	1	1	1	0	1
1	0	0	0	0	1
1	0	0	1	0	1
1	0	1	0	1	0
1	0	1	1	0	1
1	1	0	0	1	0
1	1	0	1	1	0
1	1	1	0	1	0
1	1	1	1	1	0

$I_3 I_4 \backslash I_1 I_2$	00	01	11	10
00	1		1	
01		1	1	
11			1	
10	1	1	1	1

$$F_1 = I_1 I_2 + I_3 \bar{I}_4 + \bar{I}_1 \bar{I}_2 \bar{I}_4 + I_2 \bar{I}_3 I_4$$

$$F_2 = \bar{F}_1$$

十进制数字键盘的优先编码器

No.

DATE / /

输入: $I_0, I_1, I_2, I_3, I_4, I_5, I_6, I_7, I_8, I_9$ (I_0 优先级最高)

输出: $\overline{D_0}, \overline{D_1}, \overline{D_2}, \overline{D_3}$

真值表

I_0	I_1	I_2	I_3	I_4	I_5	I_6	I_7	I_8	I_9	$\overline{D_0}$	$\overline{D_1}$	$\overline{D_2}$	$\overline{D_3}$
0	*	x	x	x	x	x	x	x	x	1	1	1	1
1	0	x	x	x	x	x	x	x	x	0	1	1	1
1	1	0	x	x	x	x	x	x	x	1	0	1	1
1	1	1	0	x	x	x	x	x	x	0	0	1	1
1	1	1	1	0	x	x	x	x	x	1	1	0	1
1	1	1	1	1	0	x	x	x	x	0	1	0	1
1	1	1	1	1	1	0	x	x	x	1	0	0	1
1	1	1	1	1	1	1	0	x	x	0	0	0	1
1	1	1	1	1	1	1	1	0	x	1	1	1	0
1	1	1	1	1	1	1	1	1	0	0	1	1	0

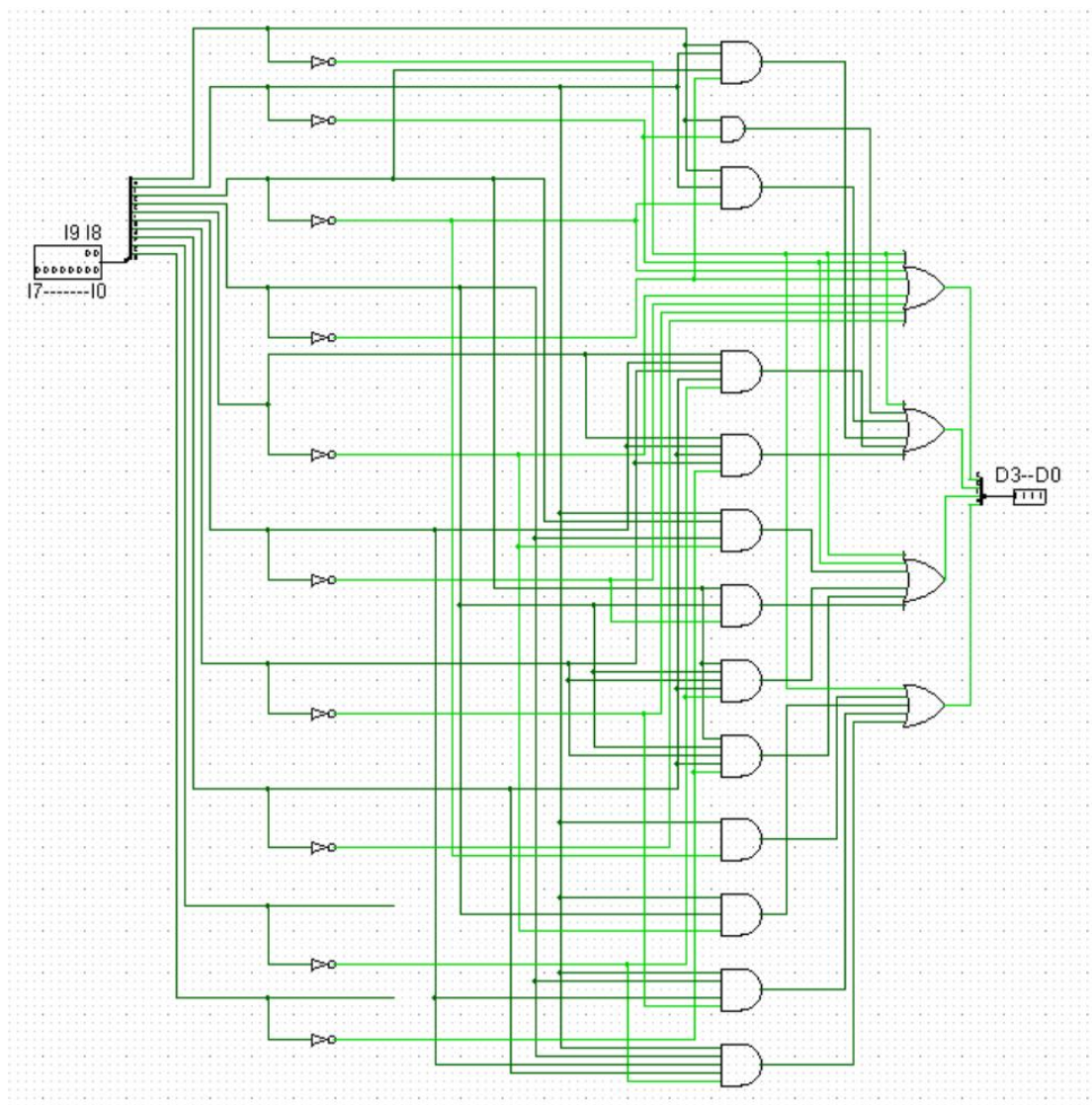
$$\overline{D_0} = \overline{I_0} + \overline{I_1} \overline{I_2} + \overline{I_1} \overline{I_3} \overline{I_4} + \overline{I_1} \overline{I_3} \overline{I_5} \overline{I_6} + \overline{I_1} \overline{I_3} \overline{I_5} \overline{I_7} \overline{I_8}$$

$$\overline{D_1} = \overline{I_0} + \overline{I_1} + \overline{I_1} \overline{I_2} \overline{I_3} \overline{I_4} + \overline{I_1} \overline{I_2} \overline{I_3} \overline{I_5} + \overline{I_1} \overline{I_2} \overline{I_3} \overline{I_6} \overline{I_7} \overline{I_8} + \overline{I_1} \overline{I_2} \overline{I_3} \overline{I_6} \overline{I_7} \overline{I_9}$$

$$\overline{D_2} = \overline{I_0} + \overline{I_0} \overline{I_1} + \overline{I_0} \overline{I_1} \overline{I_2} + \overline{I_0} \overline{I_1} \overline{I_2} \overline{I_3} + \overline{I_4} \overline{I_5} \overline{I_6} \overline{I_7} \overline{I_8} + \overline{I_4} \overline{I_5} \overline{I_6} \overline{I_7} \overline{I_9}$$

$$\overline{D_3} = \overline{I_0} + \overline{I_1} + \overline{I_2} + \overline{I_3} + \overline{I_4} + \overline{I_5} + \overline{I_6} + \overline{I_7}$$

Logicsim 原理图：



验证：

